

An *in vitro* colon model for efficient gut microbiome intervention screening: proof-of-concept using an engineered anti-TNF- α -producing *Escherichia coli*

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Supplementary methods, figures, and tables

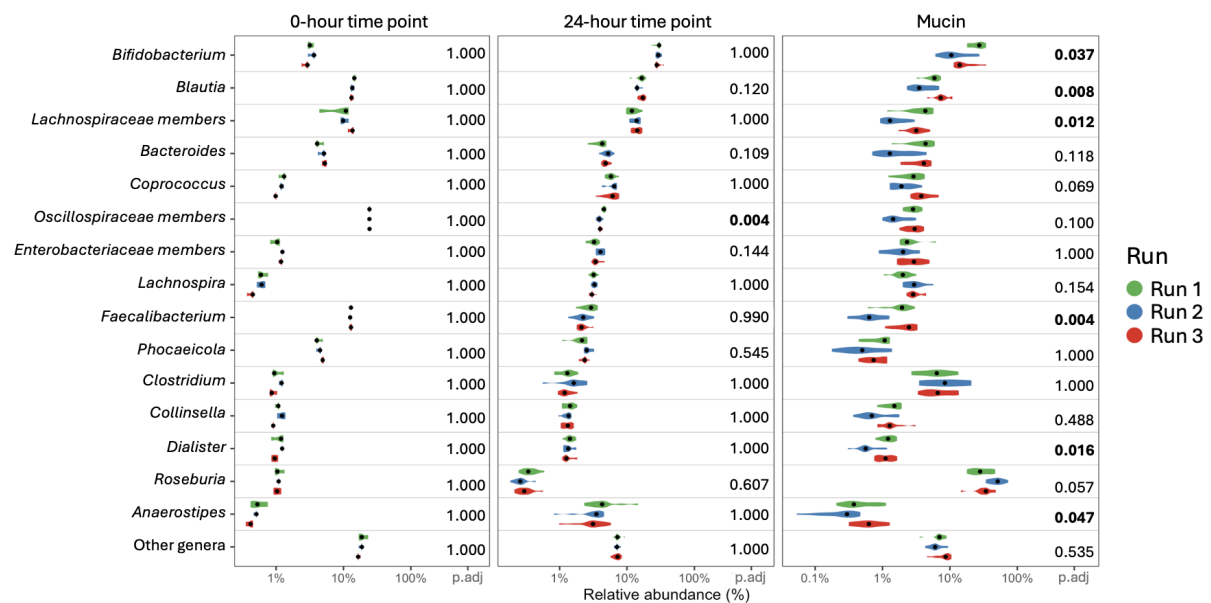
1. Supplementary methods

Supplementary description to method section 3.4 TNF- α binding ELISA setup for detection of anti-TNF- α in *in vivo* and *in vitro* samples

Clear Flat-Bottom Immuno 96-Well Plates (Catalog no. 442404, Thermo Fisher Scientific, Waltham, Massachusetts, USA) were coated with Gibco human TNF- α recombinant protein, PeproTech (Catalog no. 17831493, Thermo Fisher Scientific, Waltham, Massachusetts, USA) in PBS (5 μ g/mL) and incubated overnight at 4°C. Each well was washed with DELFIA wash buffer (Wash Concentrate in a 1:25 dilution in distilled water) (Catalog no. 1244-114, Perkin Elmer, Waltham, Massachusetts, USA). To saturate excess protein-binding sites, 150 μ L of 1% Reagent Diluent concentrate (Catalog no. DY995, Biotechne, McKinley Place Minneapolis, USA) was added to each well and incubated for one hour at room temperature. Then, each well was washed with DELFIA wash buffer. In duplicates, sample supernatants were added to the plate in 10-fold serial dilutions, parallel to a 2-fold serial dilution of a known concentration of anti-TNF- α , to generate the standard curve. This was followed by incubation for two hours (25°C, 300 rpm). After incubation, each well was washed with DELFIA wash buffer. Next, monoclonal ANTI-FLAG® M2 antibody produced in mice (Catalog no. F1804, Merck KGaA, Darmstadt, Germany) was diluted in DELFIA assay buffer (Catalog no. 1244-111, Perkin Elmer, Waltham, Massachusetts, USA) to a concentration of 0.5 μ g/mL and 100 μ L was added to each well. This was followed by incubation for one hour (25°C, 300 rpm), and afterward, each well was washed with DELFIA wash buffer. Subsequently, 100 μ L of 0.5 μ g/mL Goat Anti-mouse IgG (Catalog no. 31431, Invitrogen, Waltham, Massachusetts, USA) diluted in DELFIA assay buffer was added to each well and incubated for one to two hours (25°C, 300 rpm). Each well was then washed with DELFIA wash buffer, and 100 μ L TMB substrate (3,3',5,5'-Tetramethylbenzidine, Catalog no. DY999B, Bio-Techne, Minneapolis, Minnesota) was added to each well, followed by incubation in the dark for 10-20 minutes. The reaction was terminated by adding 50 μ L 1M H₂SO₄ (Catalog no. 1603131003, Merck KGaA, Darmstadt, Germany) to each well. Absorbance was measured at 450 nm using a plate reader, and concentrations were calculated using Microsoft Excel (2).

2. Assessment of fermentation reproducibility of the SNIPR-Gut *in vitro* colon model

2.1 Supplementary Figures related to Batch fermentation setup

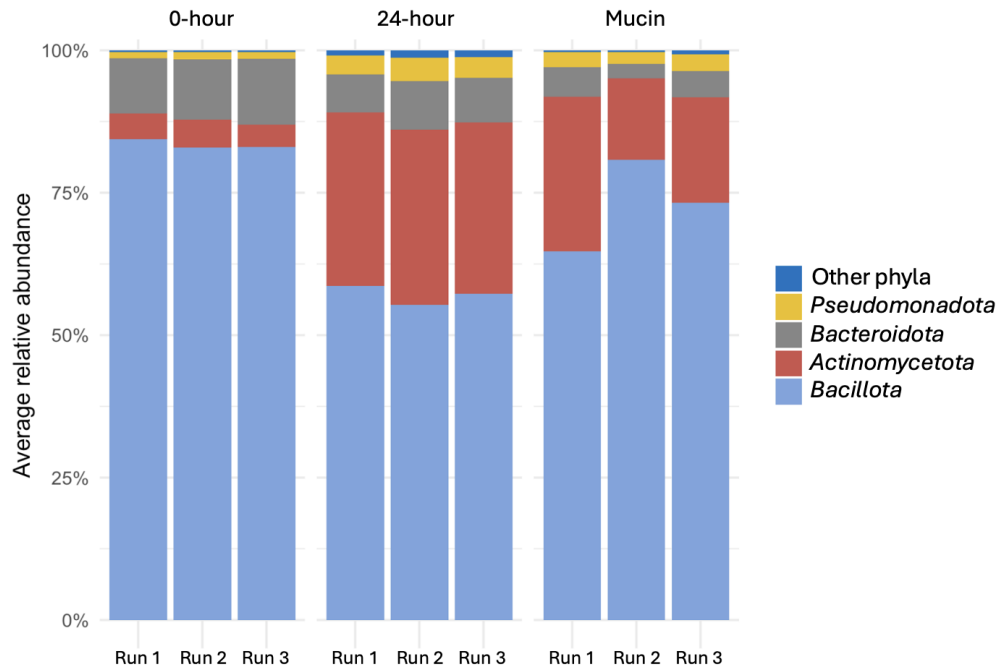


Supplementary Figure 1: Mean relative abundance of the most abundant bacteria at the genus level in the batch fermentation runs. Log-scale Rabu-plot of the top genera, collapsed by run, and grouped by time point and mucin microcosm, with mucin microcosm microbiome mapping at the endpoint of the fermentation. For each genus, Kruskal-Wallis test results comparing runs (Bonferroni-corrected across all time points (p.adj)) are presented on the right-hand panel. Samples included: 0h: n=9 (n=3 for each run); 24h: n=34, (Run 1: n=11, Run 2: n=11, Run 3: n=12); mucin microcosm: n=33, (n=11 for each run). Significant values are highlighted in bold.

Supplementary Table 1: Differences in bacterial composition between batch fermentation runs.

For each time point and mucin, Bray-Curtis and Jaccard distance matrices were used to assess differences between batch fermentation runs (Pairwise PERMANOVA, Bonferroni corrected p values (p.adj)). Samples included: Run1: n=4; Run 2: n=5; Run 3: n=3 (Except for Run1-72h: n=3 and Run2-0h: n=4.). Significant values are highlighted in bold.

Index	Time point	Comparison	F	R ²	p-value	p.adj
Bray-Curtis	0 hour	Run2 vs Run3	1.27	0.24	0.100	0.900
Bray-Curtis	0 hour	Run2 vs Run1	0.94	0.19	0.700	1.000
Bray-Curtis	0 hour	Run3 vs Run1	1.35	0.25	0.100	0.900
Bray-Curtis	24 hours	Run2 vs Run3	1.22	0.05	0.101	0.909
Bray-Curtis	24 hours	Run2 vs Run1	2.65	0.12	0.001	0.009
Bray-Curtis	24 hours	Run3 vs Run1	2.34	0.10	0.001	0.009
Bray-Curtis	Mucin	Run1 vs Run2	4.32	0.18	0.001	0.009
Bray-Curtis	Mucin	Run1 vs Run3	1.71	0.08	0.001	0.009
Bray-Curtis	Mucin	Run2 vs Run3	3.17	0.14	0.002	0.018
Jaccard	0 hour	Run2 vs Run3	1.22	0.23	0.100	0.900
Jaccard	0 hour	Run2 vs Run1	0.96	0.19	0.700	1.000
Jaccard	0 hour	Run3 vs Run1	1.27	0.24	0.100	0.900
Jaccard	24 hours	Run2 vs Run3	1.19	0.05	0.110	0.990
Jaccard	24 hours	Run2 vs Run1	2.27	0.10	0.001	0.009
Jaccard	24 hours	Run3 vs Run1	2.06	0.09	0.001	0.009
Jaccard	Mucin	Run1 vs Run2	3.31	0.14	0.001	0.009
Jaccard	Mucin	Run1 vs Run3	1.53	0.07	0.001	0.009
Jaccard	Mucin	Run2 vs Run3	2.48	0.11	0.001	0.009



Supplementary Figure 2: Phylum-level average relative abundance in the batch fermentation setup. Samples are collapsed by run and grouped by time point and mucin microcosm. Samples included: 0h: n=9 (n=3 for each run); 24h: n=34, (Run 1: n=11, Run 2: n=11, Run 3: n=12); mucin microcosm: n=33, (n=11 for each run).

Supplementary Table 2: Batch mode temporal phylum development. Wilcoxon signed-rank tests (matched pairs) with Bonferroni-corrected p-values (p.adj), were used to compare the relative abundance of the 15 most abundant phyla between adjacent time points. Significant values are highlighted in bold.

Phylum	W Statistics	Comparison	p-value	p.adj
<i>Actinomycetota</i>	45	0 vs 24	0.01	0.14
<i>Bacillota</i>	0	0 vs 24	0.01	0.14
<i>Bacteroidota</i>	1	0 vs 24	0.01	0.19
<i>Pseudomonadota</i>	45	0 vs 24	0.01	0.14
<i>Actinomycetota</i>	33	24 vs Mucin	1.0e-05	1.5E-04
<i>Bacillota</i>	543	24 vs Mucin	2.8e-06	4.3E-05
<i>Bacteroidota</i>	9	24 vs Mucin	1.3e-06	1.9E-05
<i>Pseudomonadota</i>	57	24 vs Mucin	6.8e-05	1.0E-03

Supplementary Table 3: Batch genus temporal phylum development. Wilcoxon signed-rank tests (matched pairs) with Bonferroni-corrected p-values (p.adj), were used to compare the relative abundance of the 15 most abundant genera between adjacent time points. Significant values are highlighted in bold.

Genus	W Statistics	Comparison	p-value	p.adj
<i>Anaerostipes</i>	45	0 vs 24	9e-03	0.3
<i>Bacteroides</i>	22	0 vs 24	1.00	1.0
<i>Bifidobacterium</i>	45	0 vs 24	9e-03	0.3
<i>Blautia</i>	44	0 vs 24	0.01	0.4
<i>Clostridium</i>	40	0 vs 24	0.04	1.0
<i>Collinsella</i>	45	0 vs 24	9e-03	0.3
<i>Coprococcus</i>	45	0 vs 24	9e-03	0.3
<i>Enterobacteriaceae</i> members	45	0 vs 24	9e-03	0.3
<i>Faecalibacterium</i>	0	0 vs 24	9e-03	0.3
<i>Lachnospira</i>	45	0 vs 24	9e-03	0.3
<i>Lachnospiraceae</i> members	37	0 vs 24	1e-01	1.0
<i>Mediterraneibacter</i>	29	0 vs 24	5e-01	1.0
<i>Oscillospiraceae</i> members	0	0 vs 24	9e-03	0.3
<i>Phocaeicola</i>	0	0 vs 24	9e-03	0.3
<i>Roseburia</i>	0	0 vs 24	9e-03	0.3
<i>Anaerostipes</i>	0	24 vs Mucin	6e-07	1.7e-05
<i>Bacteroides</i>	86	24 vs Mucin	5e-04	0.02
<i>Bifidobacterium</i>	31	24 vs Mucin	9e-06	2.6e-04
<i>Blautia</i>	0	24 vs Mucin	6e-07	1.7e-05
<i>Clostridium</i>	561	24 vs Mucin	6e-07	1.7e-05
<i>Collinsella</i>	205	24 vs Mucin	2e-01	1.0
<i>Coprococcus</i>	5	24 vs Mucin	9e-07	2.7e-05
<i>Enterobacteriaceae</i> members	54	24 vs Mucin	5e-05	1.6e-03
<i>Faecalibacterium</i>	101	24 vs Mucin	1e-03	0.04
<i>Lachnospira</i>	116	24 vs Mucin	3e-03	0.1
<i>Lachnospiraceae</i> members	0	24 vs Mucin	6e-07	1.7e-05
<i>Mediterraneibacter</i>	561	24 vs Mucin	6e-07	1.7e-05
<i>Oscillospiraceae</i> members	0	24 vs Mucin	6e-07	1.7e-05
<i>Phocaeicola</i>	1	24 vs Mucin	6e-07	1.9e-05
<i>Roseburia</i>	561	24 vs Mucin	6e-07	1.7e-05

Supplementary Table 4: Batch mode temporal development in the within-sample diversity metrics (Shannon, Simpson, and observed taxa). To identify differences, Dunn's test, with Bonferroni-correction, was applied (p.adj). Samples included: 0h: n=9 (n=3 for each run); 24h: n=34, (Run 1: n=11, Run 2: n=11, Run 3: n=12); mucin microcosm: n=33, (n=11 for each run). Significant values are highlighted in bold.

Metric	Comparison	Statistic	p-value	p.adj
Observed	0 vs 24	-2.67	0.008	0.023
Observed	0 vs Mucin	-6.56	5.3e-11	1.6e-10
Observed	24 vs Mucin	-6.00	1.9e-9	5.6e-9
Shannon	0 vs 24	-3.51	4.4e-4	0.001
Shannon	0 vs Mucin	-3.43	5.9e-4	0.002
Shannon	24 vs Mucin	0.10	0.920	1.000
Simpson	0 vs 24	-2.49	0.013	0.039
Simpson	0 vs Mucin	-1.47	0.142	0.427
Simpson	24 vs Mucin	1.56	0.119	0.358

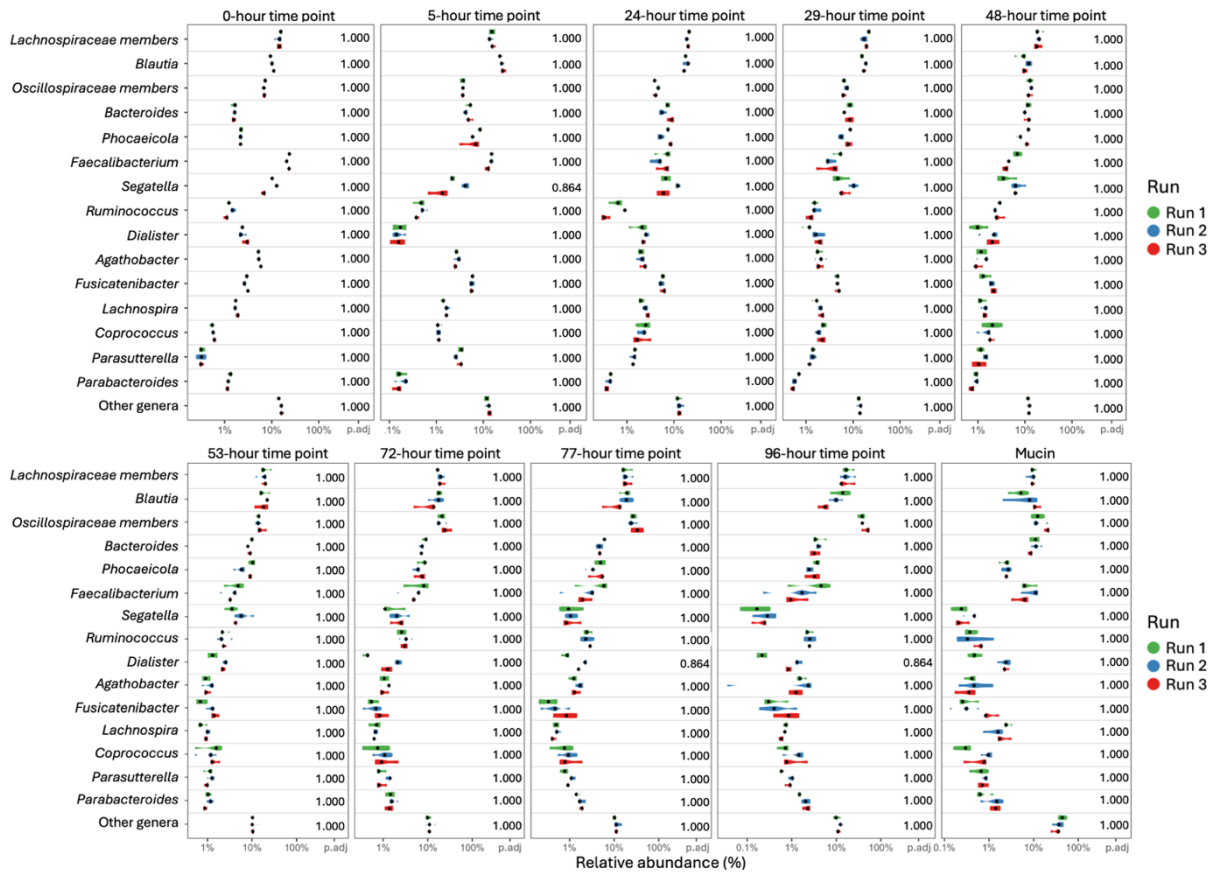
2.2 Supplementary Figures related to semi-continuous fermentation setup

Supplementary Table 5: Differences in bacterial composition between semi-continuous fermentation runs at each time point and mucin. Bray-Curtis and Jaccard distance matrices were used to assess differences between semi-continuous fermentation runs (Pairwise PERMANOVA, Bonferroni corrected p values (p.adj)). Samples for each time point and mucin microcosm included: Run1: n=4; Run 2: n=5; Run 3: n=3 (Except for Run1-72h: n=3 and Run2-0h: n=4.). Significant values are highlighted in bold.

Index	Time point	Comparison	F	R ²	p-value	p.adj
Bray-Curtis	0	Run1 vs Run2	1.36	0.18	0.076	1.00
Bray-Curtis	0	Run1 vs Run3	1.53	0.23	0.022	0.66
Bray-Curtis	0	Run2 vs Run3	1.49	0.23	0.025	0.75
Bray-Curtis	5	Run1 vs Run2	1.01	0.13	0.446	1.00
Bray-Curtis	5	Run1 vs Run3	1.39	0.22	0.028	0.84
Bray-Curtis	5	Run2 vs Run3	1.36	0.18	0.079	1.00
Bray-Curtis	24	Run1 vs Run2	0.97	0.12	0.575	1.00
Bray-Curtis	24	Run1 vs Run3	1.12	0.18	0.312	1.00
Bray-Curtis	24	Run2 vs Run3	1.899	0.24	0.021	0.63
Bray-Curtis	29	Run1 vs Run2	1.35	0.16	0.049	1.00
Bray-Curtis	29	Run1 vs Run3	1.78	0.26	0.025	0.75
Bray-Curtis	29	Run2 vs Run3	1.66	0.22	0.029	0.87
Bray-Curtis	48	Run1 vs Run2	1.58	0.18	0.012	0.36
Bray-Curtis	48	Run1 vs Run3	1.61	0.24	0.032	0.96
Bray-Curtis	48	Run2 vs Run3	1.43	0.19	0.020	0.60
Bray-Curtis	53	Run1 vs Run2	1.42	0.17	0.026	0.78
Bray-Curtis	53	Run1 vs Run3	1.41	0.22	0.034	1.00
Bray-Curtis	53	Run2 vs Run3	1.21	0.17	0.198	1.00
Bray-Curtis	72	Run1 vs Run2	1.85	0.24	0.023	0.69
Bray-Curtis	72	Run1 vs Run3	1.39	0.26	0.100	1.00
Bray-Curtis	72	Run2 vs Run3	1.37	0.19	0.031	0.93
Bray-Curtis	77	Run1 vs Run2	2.25	0.24	0.014	0.42
Bray-Curtis	77	Run1 vs Run3	1.78	0.26	0.029	0.87
Bray-Curtis	77	Run2 vs Run3	1.32	0.18	0.052	1.00
Bray-Curtis	96	Run1 vs Run2	2.02	0.22	0.011	0.33
Bray-Curtis	96	Run1 vs Run3	1.76	0.26	0.028	0.84

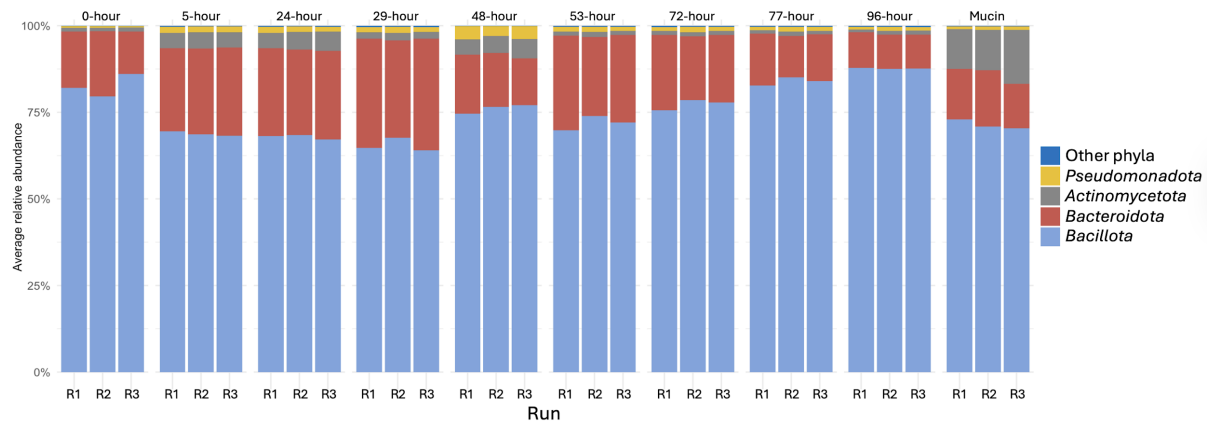
Bray-Curtis	96	Run2 vs Run3	1.28	0.18	0.112	1.00
Bray-Curtis	Mucin	Run1 vs Run2	1.00	0.13	0.492	1.00
Bray-Curtis	Mucin	Run1 vs Run3	1.63	0.25	0.040	1.00
Bray-Curtis	Mucin	Run2 vs Run3	1.04	0.15	0.384	1.00
Jaccard	0	Run1 vs Run2	1.299	0.18	0.090	1.00
Jaccard	0	Run1 vs Run3	1.44	0.22	0.030	0.90
Jaccard	0	Run2 vs Run3	1.40	0.22	0.026	0.78
Jaccard	5	Run1 vs Run2	1.01	0.13	0.448	1.00
Jaccard	5	Run1 vs Run3	1.31	0.21	0.024	0.72
Jaccard	5	Run2 vs Run3	1.27	0.17	0.083	1.00
Jaccard	24	Run1 vs Run2	1.02	0.13	0.426	1.00
Jaccard	24	Run1 vs Run3	1.098	0.18	0.330	1.00
Jaccard	24	Run2 vs Run3	1.69	0.22	0.019	0.57
Jaccard	29	Run1 vs Run2	1.28	0.15	0.039	1.00
Jaccard	29	Run1 vs Run3	1.60	0.24	0.038	1.00
Jaccard	29	Run2 vs Run3	1.50	0.20	0.039	1.00
Jaccard	48	Run1 vs Run2	1.46	0.17	0.007	0.21
Jaccard	48	Run1 vs Run3	1.47	0.23	0.030	0.90
Jaccard	48	Run2 vs Run3	1.3	0.18	0.015	0.45
Jaccard	53	Run1 vs Run2	1.34	0.16	0.018	0.54
Jaccard	53	Run1 vs Run3	1.34	0.21	0.019	0.57
Jaccard	53	Run2 vs Run3	1.17	0.16	0.260	1.00
Jaccard	72	Run1 vs Run2	1.63	0.21	0.017	0.51
Jaccard	72	Run1 vs Run3	1.299	0.25	0.100	1.00
Jaccard	72	Run2 vs Run3	1.29	0.18	0.048	1.00
Jaccard	77	Run1 vs Run2	1.96	0.22	0.010	0.30
Jaccard	77	Run1 vs Run3	1.595	0.24	0.030	0.90
Jaccard	77	Run2 vs Run3	1.26	0.17	0.042	1.00
Jaccard	96	Run1 vs Run2	1.797	0.20	0.009	0.27
Jaccard	96	Run1 vs Run3	1.60	0.24	0.025	0.75
Jaccard	96	Run2 vs Run3	1.22	0.17	0.100	1.00
Jaccard	Mucin	Run1 vs Run2	0.996	0.12	0.517	1.00
Jaccard	Mucin	Run1 vs Run3	1.48	0.23	0.029	0.87
Jaccard	Mucin	Run2 vs Run3	1.06	0.15	0.310	1.00

Supplementary Figure 3: Mean relative abundance of the most abundant bacteria at the genus level in the semi-continuous fermentation runs. Log-scale Rabu-plot of the top genera, collapsed by run, and grouped by time point and mucin microcosm, with mucin microcosm microbiome mapping at the endpoint of the fermentation. For each genus, Kruskal-Wallis test results comparing runs (Bonferroni-corrected across all time points) are presented on the right-hand panel. Samples included: Run1: n=4; Run 2: n=5; Run 3: n=3 (Except for Run1-72h: n=3 and Run2-0h: n=4.)



Supplementary Table 6: Temporal development in bacteria composition in the semi-continuous mode. Bray-Curtis dissimilarity and Jaccard distance were measured, and adjacent time points were compared (Pairwise PERMANOVA, Bonferroni corrected p values (p.adj)). Samples for each time point and mucin microcosm included: Run1: n=4; Run 2: n=5; Run 3: n=3 (Except for Run1-72h: n=3 and Run2-0h: n=4.). Significant values are highlighted in bold.

Metric	Comparison	F	R²	p-value	p.adj
Bray-Curtis	0 vs 5	11.23	0.35	0.001	0.008
Bray-Curtis	5 vs 24	9.66	0.31	0.001	0.008
Bray-Curtis	24 vs 29	1.71	0.07	0.003	0.024
Bray-Curtis	29 vs 48	5.799	0.21	0.002	0.016
Bray-Curtis	48 vs 53	1.49	0.06	0.001	0.008
Bray-Curtis	53 vs 72	2.11	0.09	0.001	0.008
Bray-Curtis	72 vs 77	1.09	0.05	0.006	0.048
Bray-Curtis	77 vs 96	1.52	0.06	0.002	0.016
Jaccard	0 vs 5	8.53	0.29	0.001	0.008
Jaccard	5 vs 24	7.35	0.25	0.001	0.008
Jaccard	24 vs 29	1.56	0.07	0.003	0.024
Jaccard	29 vs 48	4.62	0.17	0.002	0.016
Jaccard	48 vs 53	1.39	0.06	0.002	0.016
Jaccard	53 vs 72	1.86	0.08	0.001	0.008
Jaccard	72 vs 77	1.06	0.05	0.008	0.064
Jaccard	77 vs 96	1.41	0.06	0.001	0.008



Supplementary Figure 4: Relative abundance of the most abundant bacteria at phylum level in the semi-continuous mode. Samples are collapsed by run, and grouped by time point, and mucin microcosm. Samples included Run1: n=4; Run 2: n=5; Run 3: n=3 (Except for Run1-72h: n=3 and Run2-0h: n=4.)

Supplementary Table 7: Temporal development of relative mean abundance at genus-level in the semi-continuous mode. Wilcoxon signed-rank tests (matched pairs) with Bonferroni-corrected p-values (p.adj), were used to compare the relative abundance of the 15 most abundant phyla between adjacent time points. Significant values are highlighted in bold.

Phylum	W Statistics	Comparison	p-value	p.adj
<i>Actinomycetota</i>	66	5 vs 0	0.004	0.31
<i>Bacillota</i>	0	5 vs 0	0.004	0.31
<i>Bacteroidota</i>	26	5 vs 0	0.563	1.00
<i>Pseudomonadota</i>	66	5 vs 0	0.004	0.31
<i>Actinomycetota</i>	31	24 vs 5	0.556	1.00
<i>Bacillota</i>	0	24 vs 5	0.003	0.20
<i>Bacteroidota</i>	78	24 vs 5	0.003	0.20
<i>Pseudomonadota</i>	0	24 vs 5	0.003	0.20
<i>Actinomycetota</i>	53	29 vs 24	0.290	1.00
<i>Bacillota</i>	22	29 vs 24	0.196	1.00
<i>Bacteroidota</i>	50	29 vs 24	0.410	1.00
<i>Pseudomonadota</i>	17	29 vs 24	0.092	1.00
<i>Actinomycetota</i>	0	48 vs 29	0.003	0.20
<i>Bacillota</i>	4	48 vs 29	0.007	0.54
<i>Bacteroidota</i>	78	48 vs 29	0.003	0.20
<i>Pseudomonadota</i>	37	48 vs 29	0.906	1.00
<i>Actinomycetota</i>	0	53 vs 48	0.003	0.20
<i>Bacillota</i>	78	53 vs 48	0.003	0.20
<i>Bacteroidota</i>	0	53 vs 48	0.003	0.20
<i>Pseudomonadota</i>	10	53 vs 48	0.025	1.00
<i>Actinomycetota</i>	21	72 vs 53	0.307	1.00
<i>Bacillota</i>	66	72 vs 53	0.004	0.31
<i>Bacteroidota</i>	0	72 vs 53	0.004	0.31
<i>Pseudomonadota</i>	24	72 vs 53	0.450	1.00
<i>Actinomycetota</i>	24	77 vs 72	0.450	1.00
<i>Bacillota</i>	66	77 vs 72	0.004	0.31
<i>Bacteroidota</i>	0	77 vs 72	0.004	0.31
<i>Pseudomonadota</i>	15	77 vs 72	0.120	1.00
<i>Actinomycetota</i>	23	96 vs 77	0.224	1.00
<i>Bacillota</i>	78	96 vs 77	0.003	0.20

<i>Bacteroidota</i>	0	96 vs 77	0.003	0.20
<i>Pseudomonadota</i>	2	96 vs 77	0.004	0.34
<i>Actinomycetota</i>	78	Mucin vs 96	0.003	0.20
<i>Bacillota</i>	0	Mucin vs 96	0.003	0.20
<i>Bacteroidota</i>	78	Mucin vs 96	0.003	0.20
<i>Pseudomonadota</i>	24	Mucin vs 96	0.255	1.00

Supplementary Table 8: Temporal development of relative mean abundance at genus-level in the semi-continuous mode. Wilcoxon signed-rank tests (matched pairs) with Bonferroni-corrected p-values (p.adj), were used to compare the relative abundance of the 15 most abundant genera between adjacent time points. Significant values are highlighted in bold.

Genus	W Statistic	Comparison	p-value	p.adj
<i>Agathobacter</i>	0	5 vs 0	0.004	0.52
<i>Bacteroides</i>	66	5 vs 0	0.004	0.52
<i>Bifidobacterium</i>	65	5 vs 0	0.005	0.69
<i>Blautia</i>	66	5 vs 0	0.004	0.52
<i>Collinsella</i>	66	5 vs 0	0.004	0.52
<i>Dialister</i>	0	5 vs 0	0.004	0.52
<i>Faecalibacterium</i>	0	5 vs 0	0.004	0.52
<i>Fusicatenibacter</i>	66	5 vs 0	0.004	0.52
<i>Lachnospira</i>	11	5 vs 0	0.056	1.00
<i>Lachnospiraceae</i> members	34	5 vs 0	0.965	1.00
<i>Oscillospiraceae</i> members	0	5 vs 0	0.004	0.52
<i>Phocaeicola</i>	66	5 vs 0	0.004	0.52
<i>Roseburia</i>	0	5 vs 0	0.004	0.52
<i>Ruminococcus</i>	0	5 vs 0	0.004	0.52
<i>Segatella</i>	0	5 vs 0	0.004	0.52
<i>Agathobacter</i>	0	24 vs 5	0.003	0.34
<i>Bacteroides</i>	78	24 vs 5	0.003	0.34
<i>Bifidobacterium</i>	59	24 vs 5	0.126	1.00
<i>Blautia</i>	0	24 vs 5	0.003	0.34
<i>Collinsella</i>	30	24 vs 5	0.505	1.00
<i>Dialister</i>	78	24 vs 5	0.003	0.34
<i>Faecalibacterium</i>	0	24 vs 5	0.003	0.34
<i>Fusicatenibacter</i>	34	24 vs 5	0.724	1.00
<i>Lachnospira</i>	78	24 vs 5	0.003	0.34
<i>Lachnospiraceae</i> members	78	24 vs 5	0.003	0.34
<i>Oscillospiraceae</i> members	72	24 vs 5	0.011	1.00
<i>Phocaeicola</i>	26	24 vs 5	0.327	1.00
<i>Roseburia</i>	1	24 vs 5	0.003	0.44
<i>Ruminococcus</i>	67	24 vs 5	0.031	1.00
<i>Segatella</i>	78	24 vs 5	0.003	0.34

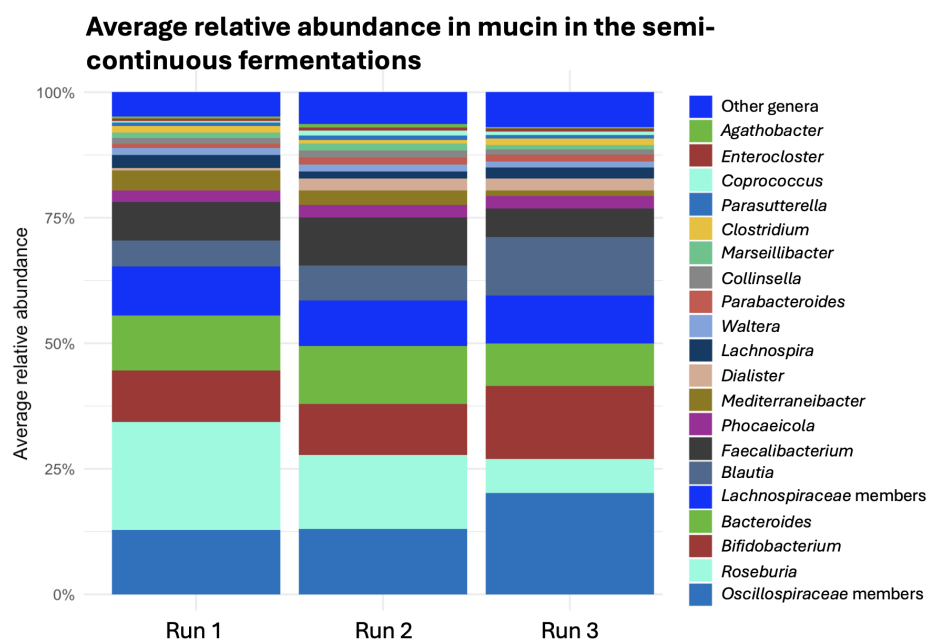
<i>Agathobacter</i>	22	29 vs 24	0.196	1.00
<i>Bacteroides</i>	70	29 vs 24	0.017	1.00
<i>Bifidobacterium</i>	59	29 vs 24	0.126	1.00
<i>Blautia</i>	36	29 vs 24	0.845	1.00
<i>Collinsella</i>	51	29 vs 24	0.367	1.00
<i>Dialister</i>	3	29 vs 24	0.005	0.72
<i>Faecalibacterium</i>	0	29 vs 24	0.003	0.34
<i>Fusicatenibacter</i>	1	29 vs 24	0.003	0.44
<i>Lachnospira</i>	0	29 vs 24	0.003	0.34
<i>Lachnospiraceae</i> members	39	29 vs 24	1.000	1.00
<i>Oscillospiraceae</i> members	78	29 vs 24	0.003	0.34
<i>Phocaeicola</i>	61	29 vs 24	0.092	1.00
<i>Roseburia</i>	70	29 vs 24	0.017	1.00
<i>Ruminococcus</i>	78	29 vs 24	0.003	0.34
<i>Segatella</i>	13	29 vs 24	0.045	1.00
<i>Agathobacter</i>	0	48 vs 29	0.003	0.34
<i>Bacteroides</i>	78	48 vs 29	0.003	0.34
<i>Bifidobacterium</i>	66	48 vs 29	0.038	1.00
<i>Blautia</i>	0	48 vs 29	0.003	0.34
<i>Collinsella</i>	0	48 vs 29	0.003	0.34
<i>Dialister</i>	52	48 vs 29	0.327	1.00
<i>Faecalibacterium</i>	73	48 vs 29	0.009	1.00
<i>Fusicatenibacter</i>	0	48 vs 29	0.003	0.34
<i>Lachnospira</i>	0	48 vs 29	0.003	0.34
<i>Lachnospiraceae</i> members	38	48 vs 29	0.969	1.00
<i>Oscillospiraceae</i> members	78	48 vs 29	0.003	0.34
<i>Phocaeicola</i>	78	48 vs 29	0.003	0.34
<i>Roseburia</i>	9	48 vs 29	0.021	1.00
<i>Ruminococcus</i>	78	48 vs 29	0.003	0.34
<i>Segatella</i>	6	48 vs 29	0.011	1.00
<i>Agathobacter</i>	2	53 vs 48	0.004	0.57
<i>Bacteroides</i>	0	53 vs 48	0.003	0.34
<i>Bifidobacterium</i>	15	53 vs 48	0.065	1.00
<i>Blautia</i>	78	53 vs 48	0.003	0.34
<i>Collinsella</i>	0	53 vs 48	0.003	0.34
<i>Dialister</i>	65	53 vs 48	0.045	1.00

<i>Faecalibacterium</i>	0	53 vs 48	0.003	0.34
<i>Fusicatenibacter</i>	0	53 vs 48	0.003	0.34
<i>Lachnospira</i>	0	53 vs 48	0.003	0.34
<i>Lachnospiraceae</i> members	26	53 vs 48	0.327	1.00
<i>Oscillospiraceae</i> members	69	53 vs 48	0.021	1.00
<i>Phocaeicola</i>	0	53 vs 48	0.003	0.34
<i>Roseburia</i>	54	53 vs 48	0.255	1.00
<i>Ruminococcus</i>	16	53 vs 48	0.078	1.00
<i>Segatella</i>	10	53 vs 48	0.025	1.00
<i>Agathobacter</i>	57	72 vs 53	0.037	1.00
<i>Bacteroides</i>	2	72 vs 53	0.007	0.90
<i>Bifidobacterium</i>	46	72 vs 53	0.266	1.00
<i>Blautia</i>	8	72 vs 53	0.029	1.00
<i>Collinsella</i>	4	72 vs 53	0.011	1.00
<i>Dialister</i>	0	72 vs 53	0.004	0.52
<i>Faecalibacterium</i>	66	72 vs 53	0.004	0.52
<i>Fusicatenibacter</i>	0	72 vs 53	0.004	0.52
<i>Lachnospira</i>	2	72 vs 53	0.007	0.90
<i>Lachnospiraceae</i> members	40	72 vs 53	0.563	1.00
<i>Oscillospiraceae</i> members	66	72 vs 53	0.004	0.52
<i>Phocaeicola</i>	8	72 vs 53	0.029	1.00
<i>Roseburia</i>	1	72 vs 53	0.005	0.69
<i>Ruminococcus</i>	62	72 vs 53	0.011	1.00
<i>Segatella</i>	0	72 vs 53	0.004	0.52
<i>Agathobacter</i>	66	77 vs 72	0.004	0.52
<i>Bacteroides</i>	0	77 vs 72	0.004	0.52
<i>Bifidobacterium</i>	27	77 vs 72	0.625	1.00
<i>Blautia</i>	65	77 vs 72	0.005	0.69
<i>Collinsella</i>	22	77 vs 72	0.351	1.00
<i>Dialister</i>	52	77 vs 72	0.100	1.00
<i>Faecalibacterium</i>	0	77 vs 72	0.004	0.52
<i>Fusicatenibacter</i>	11	77 vs 72	0.056	1.00
<i>Lachnospira</i>	0	77 vs 72	0.004	0.52
<i>Lachnospiraceae</i> members	20	77 vs 72	0.266	1.00
<i>Oscillospiraceae</i> members	66	77 vs 72	0.004	0.52
<i>Phocaeicola</i>	0	77 vs 72	0.004	0.52

<i>Roseburia</i>	60	77 vs 72	0.018	1.00
<i>Ruminococcus</i>	11	77 vs 72	0.056	1.00
<i>Segatella</i>	1	77 vs 72	0.005	0.69
<i>Agathobacter</i>	56	96 vs 77	0.196	1.00
<i>Bacteroides</i>	0	96 vs 77	0.003	0.34
<i>Bifidobacterium</i>	22	96 vs 77	0.196	1.00
<i>Blautia</i>	1	96 vs 77	0.003	0.44
<i>Collinsella</i>	16	96 vs 77	0.078	1.00
<i>Dialister</i>	0	96 vs 77	0.003	0.34
<i>Faecalibacterium</i>	10	96 vs 77	0.025	1.00
<i>Fusicatenibacter</i>	46	96 vs 77	0.610	1.00
<i>Lachnospira</i>	78	96 vs 77	0.003	0.34
<i>Lachnospiraceae</i> members	20	96 vs 77	0.147	1.00
<i>Oscillospiraceae</i> members	78	96 vs 77	0.003	0.34
<i>Phocaeicola</i>	0	96 vs 77	0.003	0.34
<i>Roseburia</i>	11	96 vs 77	0.031	1.00
<i>Ruminococcus</i>	36	96 vs 77	0.845	1.00
<i>Segatella</i>	0	96 vs 77	0.003	0.34
<i>Agathobacter</i>	2	Mucin vs 96	0.004	0.57
<i>Bacteroides</i>	78	Mucin vs 96	0.003	0.34
<i>Bifidobacterium</i>	78	Mucin vs 96	0.003	0.34
<i>Blautia</i>	24	Mucin vs 96	0.255	1.00
<i>Collinsella</i>	78	Mucin vs 96	0.003	0.34
<i>Dialister</i>	78	Mucin vs 96	0.003	0.34
<i>Faecalibacterium</i>	77	Mucin vs 96	0.003	0.44
<i>Fusicatenibacter</i>	39	Mucin vs 96	1.000	1.00
<i>Lachnospira</i>	78	Mucin vs 96	0.003	0.34
<i>Lachnospiraceae</i> members	0	Mucin vs 96	0.003	0.34
<i>Oscillospiraceae</i> members	0	Mucin vs 96	0.003	0.34
<i>Phocaeicola</i>	15	Mucin vs 96	0.065	1.00
<i>Roseburia</i>	78	Mucin vs 96	0.003	0.34
<i>Ruminococcus</i>	0	Mucin vs 96	0.003	0.34
<i>Segatella</i>	61	Mucin vs 96	0.092	1.00

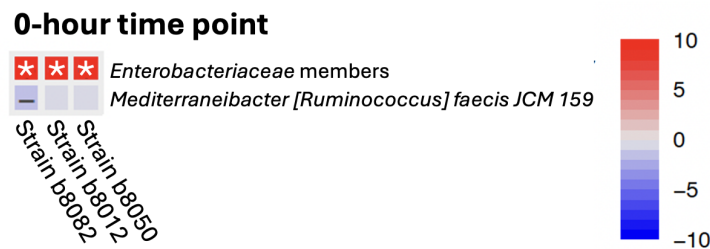
Supplementary Table 9: Temporal development in the within-sample diversity in the semi-continuous fermentation setup. Samples were rarefied to even depth and alpha diversity metrics were calculated (Shannon, Simpson, and observed taxa). To evaluate the within-sample diversity temporal development, adjacent time points were compared with, Dunn's test (Bonferroni-correction (p.adj)). Samples for each time point and mucin microcosm included: Run1: n=4; Run 2: n=5; Run 3: n=3 (Except for Run1-72h: n=3 and Run2-0h: n=4.) Significant values are highlighted in bold.

Index	Comparison	Statistic	p-value	n.pairs	p.adj
Observed	0 vs 5	66.0	0.004	11	0.030
Observed	5 vs 24	55.0	0.006	12	0.047
Observed	24 vs 29	57.0	0.006	12	0.0463
Observed	29 vs 48	2.5	0.005	12	0.037
Observed	48 vs 53	48.0	0.328	11	1.000
Observed	53 vs 72	26.0	0.722	11	1.000
Observed	72 vs 77	37.5	0.722	11	1.000
Observed	77 vs 96	64.0	0.003	12	0.023
Shannon	0 vs 5	64.0	0.007	12	0.055
Shannon	5 vs 24	20.0	0.151	12	1.000
Shannon	24 vs 29	46.0	0.623	12	1.000
Shannon	29 vs 48	73.0	0.005	12	0.039
Shannon	48 vs 53	66.0	0.005	12	0.039
Shannon	53 vs 72	64.0	0.003	12	0.023
Shannon	72 vs 77	78.0	4.9e-4	12	0.004
Shannon	77 vs 96	78.0	4.9e-4	12	0.004
Simpson	0 vs 5	11.0	0.042	12	0.336
Simpson	5 vs 24	11.0	0.027	12	0.212
Simpson	24 vs 29	37.0	0.997	12	1.000
Simpson	29 vs 48	54.0	0.266	12	1.000
Simpson	48 vs 53	68.0	0.21	12	0.168
Simpson	53 vs 72	66.0	0.016	12	0.126
Simpson	72 vs 77	78.0	7.8e-4	12	0.008
Simpson	77 vs 96	78.0	4.9e-4	12	0.004

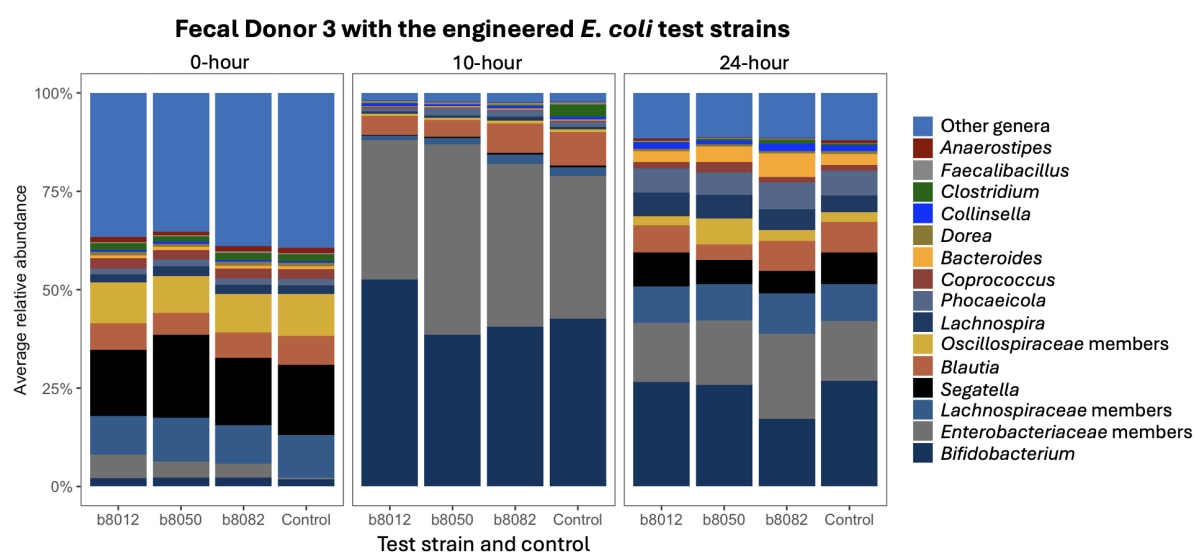


Supplementary Figure 5: Relative abundance of the most abundant bacteria at genus level in the mucin microcosms of semi-continuous fermentation setup. Because the “Other genera” group in the Figure 5D (main article) dominates in mucin microcosms, this provides a more detailed insight to the mucin colonizers 2 in the semi-continuous fermentation setup. Samples included Run1: n=4; Run 2: n=5; Run 3: n=3.

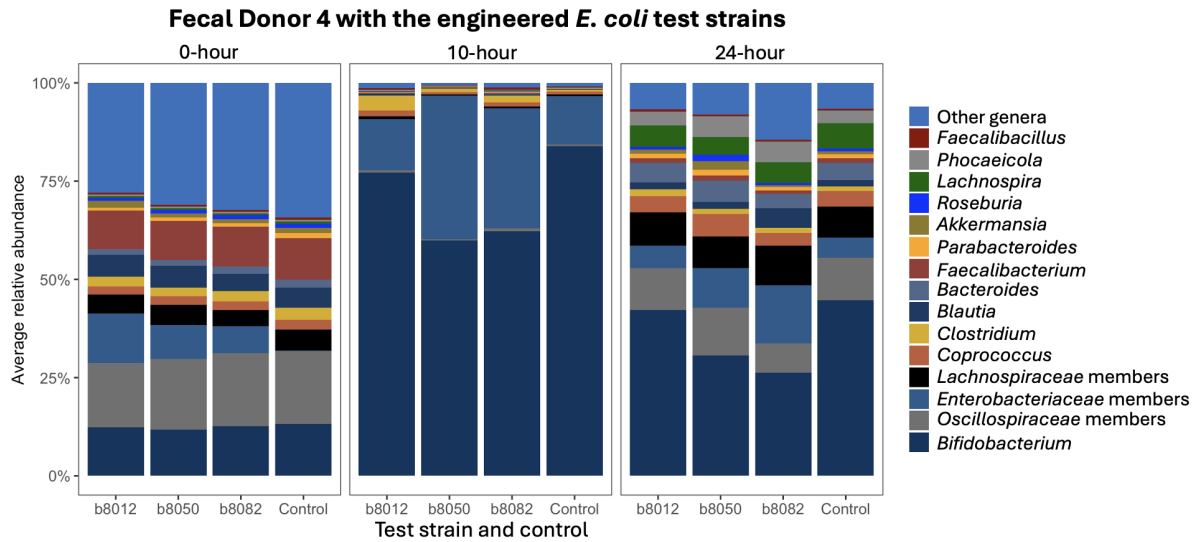
3. The influence of engineered *E. coli* strains on the *in vitro* simulated colonic microbiome



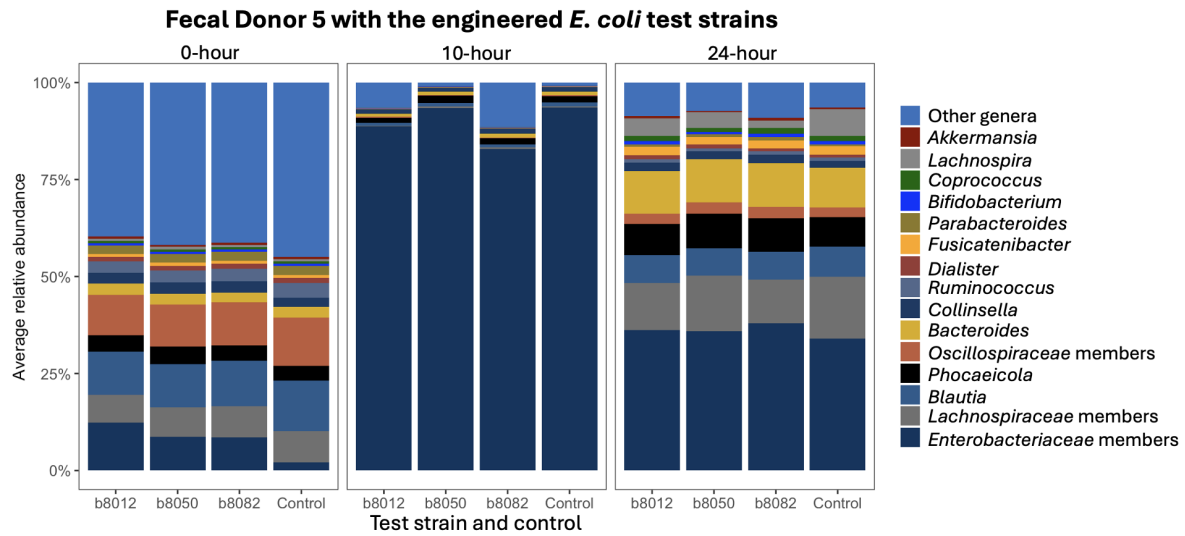
Supplementary Figure 6: The influence of engineered *E. coli* strains on *in vitro* simulated colonic microbiome composition relative to control (No addition of test strain) at the 0-hour time point. Separate linear models were built for each time point (10-, and 24-hour time point) using the MaAsLin2(1) R package, with data normalization by centered log-ratio transformation (CLR) followed by log transformation. At each time point, samples are collapsed by strain (*E. coli* b8082, b8012, and b8050) or control (no strain added), with the control as reference. To account for donor-specific variability, fecal donor (D3, D4, D5) was included as a random effect using MaAsLin2's variance partitioning, while strain and control were modelled as fixed effects. Only taxa with p-values < 0.05 for any strain are shown. The color label presents the effect size (beta coefficients): Blue(-)=negative association, read(+) = positive association, and light colors without +/- = not association. White star = significant after Benjamin Hochberg-adjustment. Each group contained 9 samples (3 donor x 3 replicates), except one missing b8050 sample (n=8).



Supplementary Figure 7: Relative mean abundance of the 15 most abundant taxa in the fecal Donor 3 fermentations. Fermentations are grouped by engineered *E. coli* test strains b8012, b8050, b8082, and control fermentation (no addition of strain) (n=3 for each group, at each time point).



Supplementary Figure 8: Relative mean abundance of the 15 most abundant taxa in the fecal Donor 4 fermentations. Fermentations are grouped by engineered *E. coli* test strains b8012, b8050, b8082, and control fermentation (no addition of strain) (n=3 for each group, at each time point, except 0-hour-control (n=2) and 10-hour-b8050 (n=2))



Supplementary Figure 9: Relative mean abundance of the 15 most abundant taxa in the fecal Donor 5 fermentations. Fermentations are grouped by engineered *E. coli* test strains b8012, b8050, b8082, and control fermentation (no addition of strain) (n=3 for each group at each time point, except 24-hour-b8082 (n=2))

References

1. Mallick H, Rahnavard A, McIver LJ, Ma S, Zhang Y, Nguyen LH, et al. MaAsLin2: Multivariable association discovery in population-scale meta-omics studies. *PLOS Comput Biol*. 2021 Nov;17(11):e1009442.
2. Wakabayashi K. How to calculate ELISA assay values by EXCEL [Internet]. Available from: <https://www.elisatests.in/wp-content/uploads/2021/04/ELISAbYEXCELE-xxx.pdf>